sizeof

→ only operator, which is a keyword and vice versa

→ looks like function, not a function

→ Syntax: sizeof(constant); variable datatype

Default real datatype is double

```
1. int main()   {
        int n=0;
        char c='c';
        float f=34.5;
        n=sizeof(int);
        printf("%d\n",n); //4
        printf("%lu\n",sizeof c); //1
*       printf("%lu\n", sizeof 'c'); //4 or 2
        printf("%lu\n", sizeof 34.5); //8
        printf("%lu\n", sizeof(float)); // 4
        printf("%lu\n",sizeof("float")); // 6
        printf("%lu\n",sizeof("12.34")); //6
        printf("%lu\n",sizeof("c")); //2
                }
```

n = sizeof(12);

n = sizeof 12;

n = sizeof int; ✗

= sizeof(int); ✓

```
2.  int main()
        {
            int i1=5, i2=2;
            int i3;
             float f;
            i3 =  i1/i2;
                   2
            f  =  i1/i2;
               =  2
            printf("i3=%d f=%f\n",i3,f); //2, 2.0
            i3=i1/(float)i2; // (float) i1 / (float) i2;
                 2.5                    2.5
            f = i1 / (float)i2;
                 2.5
           printf("i3=%d f=%f\n",i3,f); //2, 2.5
}
```

i3 = i1/i2
int/int

f =

$2\overline{)5}(2$
4
1

$2.0\overline{)5.0}(2.5$
40
10
10
0

Precedence & Associativity

```
3.  int main()
{
          40    40    40
        int x=10,y=20,z=30;
        printf("%d %d %d\n",x,y,z);
        x=y=z=40;
        printf("%d %d %d\n",x,y,z);
}
```

~~50 = 7;~~

x = y = (50 = z = 40)

↑ Syntax Error

Relational operators: < <= > >= == !=

equality   inequality

→ Binary operators

→ results in true or false
1 0

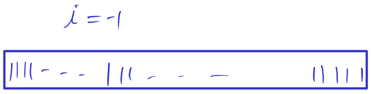
```
4. int main(){
        int x=10,y=20,z=30,result=-1;
        result=z>y>x; // r=((z>y)>x);
        printf("result=%d\n",result);   0
        }
```

```
5. int main(){
            int i=20.5;   // i=20
            float f=20.4;  // f=20.4
            int result;
            result= i==f; =>   (float)i==f  => 0
            result=i==(int)f;   i == (int)f => 1
            result= (float)i==f;    20     20
            result=i<f;    20.0<20.4 // 1
            result=i<(int)f;   20 < 20 // 0
            }
```

same

```
6.  int main(){
            int  i=-1;
            int result;        result = (int) sizeof(i) > i;
            result= sizeof(i) > i ;        4  > -1
            printf("result=%d\n",result);  // 0     1
            }
```

i=-1



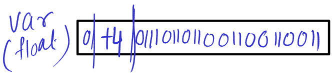
sizeof(i)
(unsigned long int) > i
signed int
⇓
sizeof(i) > (unsigned long int) i
4   > 4.3 bn

```
7. int main(){
            int a=~~10~~ 20,b=~~20~~ 10;
            printf("a=%d b=%d\n",a,b);  10, 20
            ~~//b=a*b/a=b;~~ => b=((a*b)/a)=b // Error
        1)  b=a*b/(a=b);  //partially Grouped. NOT valid for translator
        2)  //b=(a*b)/(a=b);     SAME
        3)  //b=((a*b)/(a=b));   200/20 => 10
            printf("a=%d b=%d\n",a,b);  20, 10
        }
```

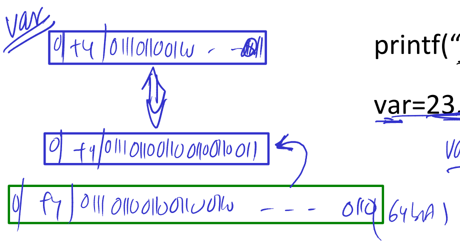
```
int x=5;
float y=2.0;
int z;
z= x%y; //Error
Pf("%d", z);
```

z = x%(int)y; ✓

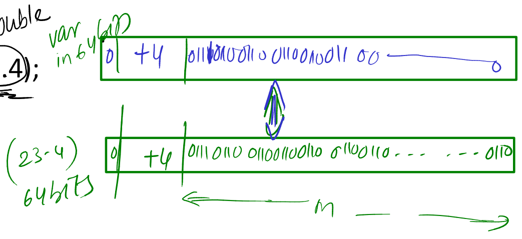
.MUL -,-
=
MOV -,-



```
8. int main(){      float var=23.4;
                    printf("%d\n",var==23.4);   double   0
                    var=23.25;                  var == 23.4f // 1
```

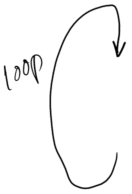


var
0 | +4 | 0111011001ω - - 011
0 | +4 | 01110110011001001011
0 | +4 | 0111 0110011001100110 - - - 0110 (64 bit)



```
        printf("%d\n", var==23.25);        }
                        1

9. int main(){
        int a=~~10~~ 11,b=~~11~~ 10;
        printf("%d\n", ++a > b--); //0        INC -   DEC -
        printf("a=%d b=%d\n",a,b); //11,10
        a=~~10~~ 11;b=~~11~~ 10;
        printf("%d\n",a++ < --b);  //0
                      10 < 10
                         0
        }

10. int main(){    int a=5,b=0,c=-3,d=-1;
        printf("%d %d %d %d\n",a,b,c,d);
        d=a&&b;
        printf("%d %d %d %d\n",a,b,c,d);
        d=b||c;
        printf("%d %d %d %d\n",a,b,c,d);
        d= !a || !c;
        printf("%d %d %d %d\n",a,b,c,d);}

11.  int main() {
        int a=5,b=0,c=-3,d=-1;
        printf("%d %d %d %d\n",a,b,c,d);
        d= ++a || ++b || ++c;
printf("%d %d %d %d\n",a,b,c,d);

        a=5;b=0;c=-3;d=-1;
        d=  b++ &&  c++ && a++;
printf("%d %d %d %d\n",a,b,c,d);

        a=5;b=0;c=-3;d=-1;
        d=  ++b &&  ++c && ++a;
printf("%d %d %d %d\n",a,b,c,d);}

12.

int main(){
```

```
sizeof(operand);        char arr[1000000];
                        sizeof(arr);

int v=~~5~~ ~~6~~ 7; float f=~~10.5~~ ~~9.5~~ 8.5;

pf("%d", v); //5                 pf("%f", f); //10.5
++v ; // Pre-Increment           --f; // Pre-dec
pf("%d", v); //6                 pf("%f", f); //9.5
v++; //Post-Increment            f--; // Post-decrement
pf("%d", v); //7                 pf("%f", f); //8.5
```



INC -

DEC -

```
var = var+1;        ADD -, -
                    MOV -, -
++var;

                    var = 5+1; ✓
                    ++5; // Wrong Syntax
a=~~10~~ 11, b=~~0~~ 11         x=~~10~~ 11, y=~~0~~ 10

b= ++a;                         y= x++;
INC, -                          MOV -, -
MOV -, -                        INC -

int ~~a~~ -              d= ++a ~~+ ++a~~ + a++ + a++;
                         pf("%d ~~%d %d %d~~", ++a, a++, ++a, a++);
```

```
int a,b,c;
int result;
printf(“enter 3 integers:”);
scanf(“%d%d%d”, &a,&b,&c);
result= (a<0) &&(b<0)&&(c<0);
result=(a>b)&&(b>c);}
```

13.

```
int   main(){          int result;
                       char ch;
                       printf(“enter a char:”);
                       scanf(“%c”,&ch);
                       result=(ch>=65)&&(ch<=90);
                       result=(((ch>=’A’)&&(ch<=’Z’)) || ((ch>=’a’)  && (ch<=’z’)))}
```

mon, tue ; wed : Parallel on Bitwise operator